

Immersive 3D Digital Twin for Collaborative Hydrogen Pipeline Simulation and Visualization: A Project Description

Muskan Sarvesh*
Department of Electrical and
Software Engineering
University of Calgary

Ryan Kang†
Department of Electrical and
Software Engineering
University of Calgary

Mehdi Marzban‡
Department of Electrical and
Software Engineering
University of Calgary

Isaac Cho§
Department of Computer Science
Utah State University

Simon Park¶
Department of Mechanical and
Manufacturing Engineering
University of Calgary

Ron Hugo||
Department of Mechanical and
Manufacturing Engineering
University of Calgary

Kangsoo Kim**
Department of Electrical and
Software Engineering
University of Calgary



Figure 1: Concept images of the proposed features in our 3D VR-based pipeline engineering tool: (a) immersive visualizations of the pipeline data will be provided for comprehension, (b) multi-user can visit the pipeline sites remotely through the virtual environment, and (c) the real-time data from the physical pipeline in our lab space will be synchronized into the virtual digital twin pipelines.

ABSTRACT

This paper introduces our plan to develop a 3D pipeline engineering tool, building upon our in-house web-based Pipeline Reliability and Integrity Management Engineering tool (PRIME), to address the limitations of traditional 2D data visualization in pipeline management. Leveraging immersive technologies, such as Virtual Reality (VR), the tool enhances simulation capabilities, introducing features like multi-user communication, digital twin integration, and virtual site visits. The study proposed in the paper aims to evaluate the tool's potential by comparing it to the existing web application, focusing on immersive training, collaboration tools, 3D data visualization, and environmental conditions in the pipeline maintenance workflows. This research aspires to contribute to the field of immersive technologies in pipeline operations, offering a comprehensive understanding of their impact on efficiency and impactful decision-making.

Index Terms: Computing methodologies—Modeling and simulation—Simulation support systems—Simulation environments; Human-centered computing—Visualization—Visualization application domains—Geographic visualization Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

*e-mail: muskan.sarvesh1@ucalgary.ca

†e-mail: ryanminseok.kang@ucalgary.ca

‡e-mail: mehdi.marzban@ucalgary.ca

§e-mail: isaac.cho@usu.edu

¶e-mail: sipark@ucalgary.ca

||e-mail: hugo@ucalgary.ca

**e-mail: kangsoo.kim@ucalgary.ca (correspondence)

1 INTRODUCTION

In the gas and oil industries, pipeline simulation and management systems are essential for both ensuring efficient material flow and addressing various use cases like maintenance, decision-making, and education/training [5]. While traditional 2D data visualization applications provide certain advantages, they encounter limitations in offering a comprehensive spatial understanding of pipeline networks, thus limiting users' ability to gain a thorough perspective. This challenge has led to the exploration of immersive 3D technologies, such as virtual reality (VR), which promise to overcome these limitations by providing a more comprehensive and holistic view of the pipeline network [6], enhancing user experience and operational efficiency.

Recent technological advances, such as the merging of web-based and VR platforms, advanced 3D visualizations, digital twin integration, and collaborative multi-user telepresence, offer significant enhancements for pipeline management systems. Researchers have demonstrated the use of VR in training for pipeline construction and oil camp operations [1, 2]. Bu et al. highlighted digital twins in valve control, using Computational Fluid Dynamics (CFD) for real-time data, improving safety, efficiency, and enabling predictive maintenance [3]. Shekargoftar et al. proposed a BIM-GIS-AR framework for better underground utility management, enhancing data collection, communication, and construction training. Santos et al. even described a Collaborative Engineering Environment (CEE) using VR in offshore engineering to bolster communication, training, and interpretation skills in complex industrial tasks [4].

Given the potential benefits and recent trends in 3D immersive technologies and digital twins, this paper outlines our approach to incorporating immersive 3D technologies into pipeline simulation and visualization tools. We aim to examine their impact on understanding pipeline states and enhancing training and education. Our main research question investigates the effects of immersive training, collaborative tools, VR-based data visualization, and risk analysis incorporating environmental factors, such as terrain and weather on

users' learning efficiency and decision-making. This study highlights the crucial interaction between technological advancements and their practical applications, potentially transforming the future of pipeline management.

2 PROPOSED 3D VR PIPELINE ENGINEERING TOOL

This section describes our proposed 3D pipeline engineering tool, which is an extension of the existing web-based pipeline visualization and analysis tool called Pipeline Reliability and Integrity Management Engineering Tool (PRIME), a tool capable of running simulations and visualizing leak detection, stress analysis, and risk assessment (Figure 2). The proposed 3D tool will include the existing functionalities of PRIME (e.g., leak detection and stress analysis simulations) while introducing an immersive VR environment with spatial visualization. This enhancement will be coupled with features, such as multi-user communication, integration with digital twins of physical pipelines, and virtual on-site visits. These additions aim to provide insights into pipeline operations and simulations and facilitate industry education/training, contributing to more informed decision-making for pipeline management. The details of the features of the proposed 3D tool are further described below.

AI-based Leak Simulation/Detection and Stress Analysis
Our 3D pipeline engineering tool will incorporate PRIME's leak simulation module, elevating conventional 2D data visualization through immersive analytics and spatial information visualization techniques. By selecting a specific node on the pipeline and inputting the leak percentage parameter, our tool will perform the pipeline leak simulation and generate the relevant pipeline data, e.g., the pressure and flow rate of the pipeline. The data can be analyzed for leak detection and stress analysis, and the results will be visualized through informative graphical illustrations, which users can interact with in a 3D immersive VR environment. This dynamic simulation capability empowers users to interpret and analyze potential leak scenarios and their impact on the overall system more effectively.

3D GIS-based Risk Analysis The risk analysis tool utilizes advanced technologies, including Cesium¹ and ArcGIS², to create an interactive map incorporating critical data such as population density, proximity of schools and residential areas to the pipeline, water crossing information, and soil types in the pipeline vicinity. A 3D pipeline model is superimposed onto this map, ensuring precise spatial representation. The tool assesses the risk associated with potential pipeline leaks using an in-house machine learning analysis module, taking into account environmental, social, and economic factors. By providing an analysis of the causes and effects of a possible leak, the tool facilitates informed decision-making for stakeholders, aiding in the development of effective risk mitigation strategies to enhance the safety and sustainability of the pipeline.

Immersive Visualization and Virtual Site-Visit The pipeline management system integrates advanced technologies, utilizing MATLAB and pump station data to offer detailed insights into node-specific pressure and flow rates within a 3D virtual environment created using the Unity game engine. Incorporating interactive elements and haptic feedback for enhanced user engagement, the system features two key scenes—the pump station and the holistic pipeline. In the pump station scene (Figure 3), a dynamic slider enables users to observe color gradients along the pipeline, representing pressure variations, while a color-coded pipeline dynamically reflects changes in gas pressure. Simultaneously, users can assess flow rates through virtual vector flux lines. The holistic pipeline scene provides real-time visualizations, including graphs illustrating



Figure 2: Pipeline leak simulation and analysis in our PRIME tool



Figure 3: Visualization of the pump station scene in VR.

overall pressure, flow rate, and anticipated temperature changes. A dynamic 3D bar graph enables status checks. This immersive system not only enhances comprehension, but also facilitates informed decision-making for pipeline maintenance and operations.

Multi-user Communication This integral component of our pipeline maintenance system facilitates real-time collaboration through features like voice chat and text messaging, enabling concurrent engagement among users, including trainers and maintenance teams. Leveraging networking libraries, e.g., PUN³, ensures a seamless and efficient collaborative platform. The adoption of a client-to-server architecture guarantees high-speed communication, enhancing the platform's interactivity. This practical approach not only optimizes collaborative communication for better decision making, but also enhances the overall training experience within the pipeline operations area.

3D Digital Twin in VR We introduce an on-campus pipeline research facility, the Advanced Pipeline Research and Innovation Laboratory (APRIL), which is a digital twin system seamlessly integrated with our web-based and VR-based platforms. This integration acts as a bridge between the virtual and physical worlds, enhancing the realism and interactivity of our pipeline system. APRIL, achieved through scanning the University of Calgary's pipeline lab and leveraging the Open Platform Communications Unified Architecture (OPC UA), ensures smooth communication between the lab's data and the VR scene. This integration facilitates essential features like diagnostics, asset management, and monitoring, contributing to a comprehensive and interactive pipeline management experience. Our digital twin system intends to enhance realism, amplifying the overall user experience and exemplifying our commitment to advancing pipeline research and innovation.

¹Cesium CesiumJS: <https://cesium.com/cesiumjs/> (Accessed 2024-01-15)

²ESRI ArcGIS: <http://www.esri.com/software/arcgis> (Accessed 2024-01-15)

³Photon Unity Networking: <https://www.photonengine.com/pun> (Accessed 2024-01-15)

3 PLANNED STUDY

To examine the potential of our proposed tool, we plan to conduct a user study employing both the proposed 3D pipeline engineering tool and the web-based PRIME.

Participants To ensure a diverse range of expertise and experience in pipeline engineering, our participant group will include graduate students from the University of Calgary who have a background in pipeline research, as well as local pipeline experts and stakeholders from relevant industry sectors.

Study Design The study employs a within-subject design comparing two pipeline management tools below:

- **2D Web-based Tool:** Participants will engage with a 2D visualization interface, concentrating on simulations for detecting leaks, understanding pipeline stress levels, and examining data visualizations overlaid on a map, all displayed on a conventional monitor.
- **Immersive 3D VR-based Tool:** Wearing an immersive VR headset, participants will interact with 3D virtual pipeline models, navigate through geographical data, collaborate within a virtual environment, and delve into 3D data visualizations.

Participants will engage with both systems in a counterbalanced order, actively simulating a variety of pipeline data scenarios while taking into account diverse environmental conditions. These conditions include factors such as the size of nearby flowing streams and rivers, prevailing weather patterns, and the proximity of communities to pipeline facilities.

Procedure Upon participants' arrival, they will undergo an orientation session covering the usage of both the 2D web-based and 3D VR-based interface tools. As the study progresses, participants will interact with each system individually, engaging in diverse data simulations and analysis scenarios, including both individual and collaborative tasks, aiming to evaluate the tools' effectiveness across various task settings. In the web interface task, participants will perform a leak detection simulation, generate graphs, and orally discuss their experience with the investigator. In the VR scene task, participants will explore the holistic pipeline view, gaining insights into terrain data, and move on to conduct a site visit to the pump station after clicking a node on the pipeline. Following this, they will be asked to perform a VR-based leak simulation. Subsequently, participants will share insights and collaborate with the investigator's avatar in the VR space. After completing tasks with each pipeline management tool, participants will provide feedback, drawing comparisons between the two scenarios.

Measures and Hypothesis The assessment of integrated pipeline simulation technologies will employ a set of measures to provide a thorough evaluation. Objective task completion metrics, including time taken to complete the given tasks, accuracy, and completion station, will offer quantitative insights into participants' performance in tasks involving immersive training, collaboration tools, VR-based data visualization, and varying environmental conditions. User satisfaction scores, derived from post-task surveys, will complement these objective metrics by offering a subjective perspective on participants' experiences, covering aspects like effectiveness, ease of use, and overall satisfaction. Qualitative feedback obtained through post-task interviews and observations will provide in-depth insights and suggestions for improvement.

Building upon our study design and selected measures, we posit the following general hypotheses:

H1: VR-based system will provide a better user experience, with increased immersion, engagement, and overall satisfaction during tasks and collaboration compared to the traditional web-based interface.

H2: VR system will improve spatial understanding and decision-making compared to the traditional web interface.

By the end of this study, we aim to attain a thorough understanding to answer the research question regarding the impact of immersive training, collaboration tools, VR-based data visualizations, and environmental considerations, such as terrain and weather, on maintenance team efficiency and decision-making within pipeline management systems.

4 SUMMARY AND FUTURE WORK

This paper outlines the development of an advanced VR-based 3D pipeline engineering tool, extending our prior web-based application. The new tool is designed to augment existing functionalities, such as leak detection and stress analysis, by employing interactive 3D visualization techniques. These techniques aim to facilitate a more profound and effective comprehension of intricate pipeline data. In addition, the tool will integrate digital twin models, which are synchronized with real-time data from physical pipelines in our laboratory setting. This integration will be complemented by additional features, including comprehensive environmental risk analysis and an immersive, multi-user VR communication platform.

The planned study is structured to evaluate the usability of our tool in terms of its contribution to contextual understanding, the quality of the immersive training experience it provides, and its efficacy in fostering collaboration. This project underscores the significant role of immersive technologies in enhancing practices in pipeline monitoring and maintenance. We eagerly anticipate engaging in further discussions on this topic at the upcoming workshop, where we aim to gather valuable insights and feedback from industry experts and peers.

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